

NAFF — Avtonom ROP Agent

MVP spetsifikatsiya · 3 hafta · 2026-10-07 · owner chat 88938071

Mundarija
1. Loyihaning maqsadi
2. ROP Agent nima — va u chatbotdan nima bilan farq qiladi
3. MVP qamrovi (nima bor va nima yo'q)
4. Arxitektura — bitta rasm
5. 10 ta KPI (nimalarni o'lchaymiz)
6. 10 ta Alert trigger (qachon push qilamiz)
7. Avtonomlik mexanizmlari (scheduler, OODA, state, approval)
8. Ertalabki brifing namunasi (real matn)
9. Operator coaching namunasi (DB-based)
10. 3-haftalik roadmap (kun bo'yicha)
11. Guardrails va risklar
12. V2 preview (nima davom etadi)

1. Loyihaning maqsadi

Siz NAFF (phone-shop call-center) uchun 24/7 ishlaydigan avtonom Head-of-Sales (ROP) agentini qurayapsiz. U chatbot EMAS — siz "qanday ish?" deb so'rashingizni kutmaydi. Har kuni ertalab 07:00 da Telegramga brifing yuboradi; qimmatli lead 2 soat kutib qolsa yoki operator conv 20% tushsa — darhol push qiladi; har kuni 11:00/15:00/19:00 da sizga eslatma yuboradi; yakshanba 09:00 da haftalik scorecard beradi.

Natija 3 haftadan keyin: siz telefonni ochganda agent allaqachon aytib qo'ygan bo'ladi: "Kecha Oybek'ning 3 ta qimmatli lead'i 8 soat ushlab turildi; bugun 10:00 da u bilan narx-e'tirozi mavzusida 20 daq coaching qiling, mana skript."

2. ROP Agent nima — va u chatbotdan nima bilan farq qiladi

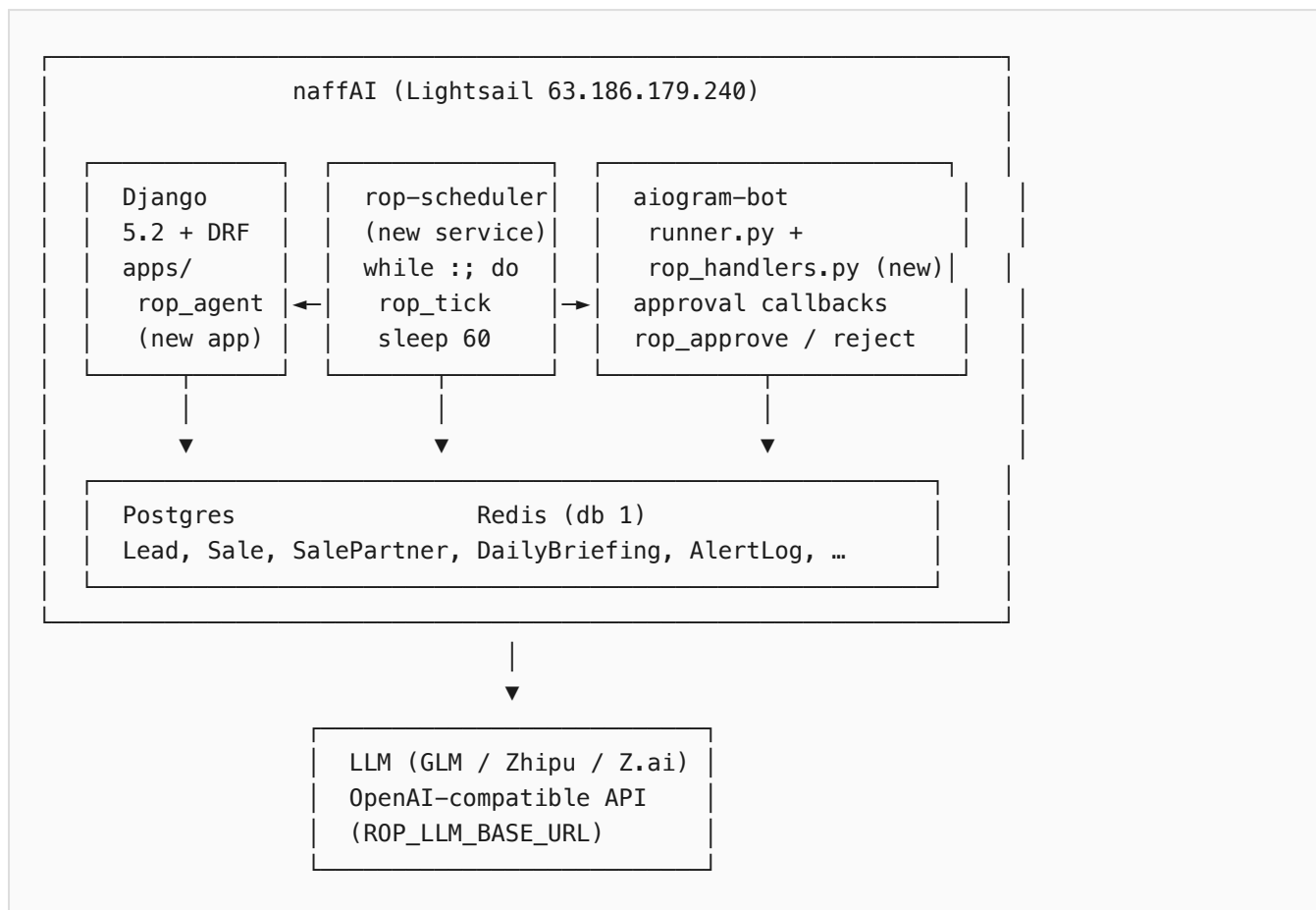
O'lcham	Chatbot	Avtonom ROP agent
Boshlash	Siz savol berasiz	Vaqt / hodisa o'zi uyg'otadi
Hodisa manbai	Chat xabar	Celery-beat yo'q — docker-compose cron + Django signal (post_save Lead/Sale/CallAttempt)
Xotira	Session	Postgres + Redis — kecha nima aytganini biladi, takrorlamaydi
Harakat	Faqat javob	Push + Reminders + Approval button + DB write (prod'da tasdiq orqali)
Rate-limit	Yo'q	Alert dedup 4h window; har alert_type uchun 10 min Redis SETNX

3. MVP qamrovi

MVP (3 hafta) ichida:

Modul	Status	Vazifa
Sales Analyst	MVP	NAFF DB'dan 10 ta KPI puller (lead/call/sale/conv/AOV/partner)
Alert Agent	MVP	10 ta threshold trigger + 4h dedup via <code>AlertLog.dedup_key</code>
ROP Brain	MVP	Synthesizer — GLM LLM orqali KPI va alert'larni brifingga aylantirish
Scheduler	MVP	Bitta <code>rop-scheduler</code> docker-compose service, har 60s <code>rop_tick</code> ni chaqiradi
Ertalabki brifing 07:00	MVP	Har kuni 07:00 Toshkentda avtomatik
Tushlik pulse 14:00 / Kechki yakun 21:00	MVP	O'z soatida avtomatik
Reminder 11/15/19	MVP	Qisqa 1-2 qatorli pulse ownerga
Approval flow (prod)	MVP	Inline button [☒ Tasdiqlayman / ☒ Yo'q], 24h expire, demo'da bypass
Weekly scorecard	MVP	Skelet yakshanba 09:00 (chuqur analitika V2'da)
YouTube Expert Watcher	V2	Keyingi faza
News Monitor UZ+xorij	V2	Keyingi faza
Learning Coach (kitob/kurs)	V2	Keyingi faza
Operator Coach (chuqur)	V2	MVP'da faqat conv-drop alert + basic weakness detect
Call Audio AI (Asterisk + Whisper)	Butunlay o'chirilgan	Pipeline qurilmaydi — foydalanuvchi qarori

4. Arxitektura — bitta rasm



Kalit qarorlar:

- **Celery YO'Q** — naffAI'da hali Celery sozlanmagan. Ishlatmaymiz. Buning o'rniga mavjud docker-compose cron pattern (`hot-leads-watch` , `morning-splitter`).
- Bitta `rop-scheduler` service — oltitta cron o'rniga. `rop_tick` har daqiqa Toshkent-local soatni tekshiradi va kerakli sub-task'ni chaqiradi.
- `apps.rop_agent` — alohida microservice emas, mavjud Django ichida. Postgres/Redis/aiogram-bot allaqachon shu yerda — tekshirish, debug oson.
- HackSoft layering: `models` → `selectors` → `services` → `apis` . ORM query faqat `selectors.py` da. LLM/TG chaqiruvlari faqat `services.py` da. Management command'lar `services.run_*`() ni chaqiradi, boshqa hech narsa.

5. 10 ta KPI

#	KPI	Formula	Qayerdan
1	leads_created	Count(Lead, created_at window)	Lead
2	leads_contacted	Count(distinct Lead with ≥ 1 CallAttempt)	CallAttempt
3	sales_count / amount / avg	Count+Sum(Sale, confirmed only)	Sale
4	conversion_rate	sales / leads_contacted	Sale, CallAttempt
5	partner_split (payment channel)	SalePartner groupby channel	SalePartner, Channel
6	qimmatli_waiting_now	Lead.hot_until & status in NEW/ASSIGNED & wait > N min	Lead
7	followup_rate_today	CallbackReminder done / total (bugun)	CallbackReminder
8	per_operator_conv_7d	Per-op: this_7d vs prev_7d	CallAttempt, Sale
9	operator_calls_today	Count(CallAttempt, operator)	CallAttempt
10	silent_operators_now	Active op & 0 call after 11:00 Tashkent	Operator, CallAttempt

Barcha selector'lar **pure read** — ORM query faqat shu yerda. HackSoft qoidasi: service'lar SQL yozmaydi, management command'lar ORM'ga tegmaydi.

6. 10 ta Alert trigger

#	Alert	Shart	Priority	Dedup
1	qimmatli_long_wait	hot_until aktiv + kutish > 2h, aloqa yo'q	RED	lead_id · 4h
2	leads_queue_overflow	leads_waiting_30min > 50	RED	global · 1h
3	operator_conv_drop	per_op delta_pp < -20	RED	op_id · 1d
4	followup_rate_low	followup_rate < 0.75	YELLOW	daily · 24h
5	partner_share_shift	share_delta > 15pp vs 7d avg	YELLOW	channel · 1d
6	big_sale_celebrate	Sale.amount ≥ 15 M so'm	INFO	sale_id · bir martalik
7	operator_silent	0 CallAttempt soat 11:00 dan keyin	RED	op_id · 1d
8	missed_qimmatli_call	no_answer + lead.hot_until	RED	lead_id · 2h
9	conv_today_slump	today_conv vs 7d_avg: delta_pp < -20	RED	daily · 1d
10	reminder_overdue_spike	CallbackReminder overdue > 30	YELLOW	daily · 4h

7. Avtonomlik mexanizmlari

7.1 OODA loop — har sub-task shu bosqichlarni bosib o'tadi

```
OBSERVE → ANALYZE → DECIDE → ACT → LEARN  
DB pull → LLM summ. → rule+LLM → TG push → state save
```

7.2 Scheduler (docker-compose, Celery YO'Q)

```
rop-scheduler:  
  build: ./backend  
  env_file: .env  
  depends_on: [db]  
  command: bash -c '  
    while ;; do  
      python manage.py rop_tick 2>&1 | tee -a /tmp/rop.log || echo "rop_tick rc=$?"  
      sleep 60  
    done'
```

`rop_tick` har daqiqa Toshkent-local soatni tekshiradi:

- **07:00** → `rop_morning_briefing`
- **14:00** → `rop_midday_pulse`
- **21:00** → `rop_evening_wrap`
- **11:00 / 15:00 / 19:00** → `rop_reminder`
- **yakshanba 09:00** → `rop_weekly_scorecard`
- **Har 10 daq (minute % 10 == 0)** → `rop_alert_check`
- **Har 15 daq (minute % 15 == 0)** → `rop_cleanup_approvals`

7.3 State va memory

Takrorlanmaslik — avtonom agent uchun eng muhim xususiyat.

- **Postgres** (uzoq xotira): `DailyBriefing` (90 kun oxirgi briefing), `AlertLog` (barcha alert history), `OperatorCoachingLog` (per-op coaching), `PendingApproval` (24h expire).
- **Redis** (qisqa xotira): `alert:<code>` SETNX bilan rate-limit, `rop:briefing_sent:today` ikki marta yuborishni to'xtatadi.
- **ROPState singleton**: `muted_until`, `last_briefing_at`, `last_pulse_at`, `last_evening_at`, `last_weekly_at`.

7.4 Approval flow (faqat prod)

- `ROP_APPROVAL_REQUIRED=true` (prod default) → har write-action `PendingApproval` yaratadi + inline button bilan push.
- Owner tasdiqlaganda: aiogram `rop_handlers.cb_rop_approve` → `services.execute_pending_approval`.

- 24h ichida javob kelmasa → **EXPIRED** (har 15 daq **rop_cleanup_approvals** ko'rib chiqadi).
- Demo (**demo.naff.flek.uz**) → **ROP_APPROVAL_REQUIRED=false** , agent avtomatik bajaradi.

7.5 Mute (sukut)

Inline-button [■ Sukut 2h] → **services.mute()** → **ROPState.muted_until = now + 2h** . Har sub-task **_is_muted()** tekshiradi. 🚫 RED alert'lar sukutni buzadi, qolgani pauza.

8. Ertalabki brifing namunasi

🇺🇿 Xayrli tong!
📅 2026-10-08

┌ KECHAGI NATIJA
| Leads: 342 | Contact: 198 | Sale: 47
| Conv: 23.7% | Revenue: 197,400,000 so'm
| Top partner: Alif
└

[LLM synthesis – Uzbek, 200–350 so'z]

┌ OPERATOR DIQQATI
| 🚫 Oybek: conv -24% kecha – ehtimol narx e'tirozida chegirma gapiryapti
| 🟢 Nilufar: 12 sale/kun – jamoada eng yaxshi
└

┌ ALERT'LAR (TOP 3)
| 📌 Qimmatli lead #4821 – Oybek'da 2h 15 min kutib turibdi
| 📌 followup_rate: 68% (ostonadan past)
| 💡 Katta sale 18.5M – Sanjar
└

┌ BUGUNGI 3 USTUVOR
| 🚫 10:00 Oybek bilan narx e'tirozi coaching (20 daq)
| 🟡 Qimmatli 7 lead'ni eng tajribali 3 operatorga qol-qo'l
| 🟢 Alif commission sync 15:00
└

Jahd o'z joyida bo'lsin – kun qisqa.

[✅ Reja qabul] [🔄 Qayta tuz] [■ Sukut 2h]

LLM-qism — Claude Sonnet emas, **GLM / Zhipu glm-4-plus**. Sarlavha ("KECHAGI NATIJA" blok, "ALERT'LAR" blok) — **services._render_briefing_header()** da rule-based (raqamlar DB'dan), LLM faqat o'rtadagi 2-3 qism uchun synthesis beradi.

9. Operator coaching namunasi (DB-based, audio YO'Q)

MVP'da Call AI yo'q. Lekin DB-based coaching baribir ishlaydi:

👤 Oybek Toshmatov — 2026-10-08

Kuchli tomoni (7 kun):

- ✅ Partner tanlashda Alif'ni to'g'ri taklif (18 sale / 20 sales)
- ✅ Ertalabki smena'da conv 28% (jamo'a avg 22%)

Zaif tomoni (7 kun):

- ❌ Umumiy conv: 14% ← -24pp vs oldingi hafta
- ❌ Tushdan keyingi smena'da sotuv 2 marta kam (DB CallAttempt vaqti analizi)
- ❌ Qimmatli lead'lar bilan ishlashda — 3 ta "no_answer" oxirgi 3 kun

🗨️ Ertaga 10:00 coaching rejasi:

1. Oxirgi 3 qimmatli no_answer sabablarini birga aniqlash
2. SPIN "Need-Payoff" savollari — 3 ta aniq variant
3. Rolik: Oybek operator, men mijoz, 3 raund
4. Vazifa: ertaga tushdan keyin 5 qo'ng'iroqda narx aytmasdan qiymat ko'rsatish
5. Yakuni 19:00 — natija

Keyingi check: 2026-10-11 (3 kun), maqsad conv $\geq$ 20%

Qanday ishlaydi: `operator_conv_drop` alert'i ishga tushgach,
`services.generate_coaching_plan(operator_id)` → selectors'dan 7 kunlik raqamlar + LLM synthesis
→ `OperatorCoachingLog` ga saqlanadi + ownerga push (prod'da approval orqali).

10. 3-haftalik roadmap

Hafta 1 — foundations (dush-juma)

1. Dush: `apps.rop_agent` skelet + 5 model + migratsiya (local SQLite'da test)
2. Sesh: Settings/env — `ROP_*` block + `pyproject.toml` openai dep
3. Chor: 10 ta KPI selector + 3 unit test
4. Pay: 10 ta alert trigger + dedup via `AlertLog.dedup_key` + 3 unit test
5. Jum: `rop_tick` scheduler + 7 ta management command (skelet)

Hafta 2 — services, LLM, TG, approval

1. Dush: `llm.py` (GLM / OpenAI-compatible) + stub fallback + routing
2. Sesh: `telegram.py` — `send_to_owner` httpx sync + inline keyboards
3. Chor: `services.py` — morning/midday/evening/reminder/alert_check
4. Pay: `PendingApproval` flow + aiogram `rop_handlers.py`
5. Jum: Prompts (NAFF context, "hamkor = payment channel" qoidasi) + integration test

Hafta 3 — demo deploy, tune, prod cutover

1. Dush: `docker-compose.prod.yml` + `rop-scheduler` + `demo-rop-scheduler` + `.env.example`

- 2. Sesh: Deploy `feat/rop-agent` → `demo.naff.flek.uz`, `ROP_ENABLED=true`, `ROP_APPROVAL_REQUIRED=false`
- 3. Chor-Pay: Live kuzatish demo'da — brifinglar, alert'lar, dedup to'g'ri ishlayaptimi
- 4. Jum: Prompt tune (brifing uzunligi, UZ tilining aniqligi), model routing (glm-4 vs glm-4-plus)
- 5. Keyingi dushanba: Prod cutover — `ROP_ENABLED=true`, `ROP_APPROVAL_REQUIRED=true`

11. Guardrails va risklar

Risk	Mitigation
LLM galyusinatsiyasi (raqam to'qib qo'yish)	Prompt'da qattiq qoida: raqamlar faqat payload'dan. Header rule-based, LLM faqat narrative.
Alert spam (storm)	Ikki qavatli dedup: <code>AlertLog.dedup_key</code> + 4h window; Redis SETNX 10 min
Prod DB'ga tasodifiy yozuv	<code>ROP_APPROVAL_REQUIRED=true</code> + <code>PendingApproval</code> har write uchun
LLM kaliti yo'q	<code>synthesize_stub</code> fallback — smoke test va TG push baribir ishlaydi
Scheduler Tashkent-time edge-case 07:00:30	$07:00 \leq \text{hhmm} < 07:01$ window + <code>DailyBriefing.date</code> unique → idempotent
Owner ta'tilda	<code>services.mute(duration)</code> — faqat RED alert'lar sukutni buzadi
Postgres down	Services <code>try/except</code> + log; <code>alert_owner</code> helper Redis down ham fallback
Telegram API down	<code>send_to_owner</code> swallow + log; <code>AlertLog</code> yozilgandan keyin push bo'ladi (keyingi tick qayta urinmaydi — qasddan)

12. V2 preview (keyingi davom)

- 1. **YouTube Expert Watcher** — Grant Cardone, Chris Voss, UZ/RU savdo kanallari. YouTube Data API v3, har 6 soatda scan, LLM 5-3-2-1 summary.
- 2. **News Monitor** — UZ (Uzum/retail/telecom) + xorij (TR/UAE/US) kanallari. Google News API + RSS + Telegram scraper.
- 3. **Learning Coach** — zaiflik (`operator_conv_drop`) → kitob/bob tanlab 7 kunlik reading plan.
- 4. **Operator Coach deep** — har hafta per-op scorecard, 6 dimension; conv-pattern analytics.
- 5. **Call AI — chetlab o'tiladi** (Asterisk integration foydalanuvchi tomonidan rad etilgan). Pipeline qurilmaydi.
- 6. **UI/Dashboard** — Django admin'dan tashqari, DRF endpoint + React page (`DailyBriefing` history, `AlertLog`, `PendingApproval`).